

1-672-338 | uomer@student.ubc.ca | [Muhammad Usman Omer](https://www.linkedin.com/in/usmanomer) 1-672-338-7943 | [uomer@student.ubc.ca](https://www.linkedin.com/in/usmanomer) | [linkedin.com/in/usmanomer](https://github.com/usmanomer1) | github.com/usmanomer1 Education University of British Columbia Vancouver, BC Bachelor of Science in Computer Science Sep. 2023 – May 2027 • Dean's List (2024) Experience Machine Learning Intern Aug 2021 – Oct 2021 Punjab Information Technology Board Lahore, Pakistan • Developed a machine learning model to track illegal license plates via CCTV using R. • Collaborated in a team to implement and test the model in a real-world setting. Machine Learning Intern Jun 2022 – Jul 2022 Himmah Capital Limited Dubai, UAE • Applied machine learning techniques to analyze trends in the Saudi Stock Exchange (TASI). • Enhanced prediction accuracy by 127% through KNN and big data analytics. Projects Ticket Booking Application | Java, Spring Boot, Spring Security, React, Vite • Designed and implemented a robust ticket booking platform with secure login and registration functionality, featuring a modern, responsive UI built with React and Vite. • Utilized Spring Data JPA to streamline CRUD operations for managing user bookings, cancellations, and updates, ensuring seamless interaction with the backend database. • Implemented Spring Security and JWT-based authentication to safeguard user data and restrict access to authorized endpoints, enhancing overall application security. • Developed RESTful API endpoints using Spring Boot REST controllers, enabling efficient communication between the frontend and backend. • Introduced a well-structured service layer to handle business logic, ensuring scalability, maintainability, and separation of concerns within the application. • Backend Repository | Frontend Repository | Live Application FUT Team Builder Application | Java, Swing, JUnit, Mockito • Developed a comprehensive application for FIFA Ultimate Team (FUT) enthusiasts to create, manage, and optimize custom teams with an interactive gaming-style interface using Java Swing. • Implemented secure login and registration functionality using Java's UserManager, ensuring role-based access control for users and admins. • Designed a complex service layer with algorithms to calculate key team metrics such as chemistry, average rating, and total price, improving the user experience. • Integrated a community feature that allows users to save and share their teams in a public repository, enabling others to browse, filter, and like popular team builds. • Achieved 90% code coverage through comprehensive unit testing using JUnit and Mockito, implementing test cases for all service layer components and user interactions. • Source Code SFTWR Labs Web Application | Java, Spring Boot, Next.js • Designed and developed a full-stack business application for SFTWR Labs, featuring a secure backend built with Spring Boot and a dynamic frontend using Next.js. • Implemented a RESTful API architecture and utilized Spring Data JPA for CRUD operations, managing data through Spring REST endpoints. • Developed an Admin Panel with Spring Security and integrated Slack API for automated notifications and Spring Mail for email communications. • Visit SFTWR Labs | Source Code (Backend) | Source Code (Frontend) Technical Skills Languages: Java, C, C++, Python SQL, JavaScript, HTML/CSS Frameworks: Spring Boot, React, Hibernate Tools: Linux, Git, Redis Cache, Amazon Bedrock, Open AI API, Apache Kafka | Muhammad Usman Omer 1-672-338-7943 | uomer@student.ubc.ca | [linkedin.com/in/usmanomer](https://www.linkedin.com/in/usmanomer) | github.com/usmanomer1 Education University of British Columbia Vancouver, BC Bachelor of Science in Computer Science Sep. 2023 – May 2027 • Dean's List (2024) Experience Machine Learning Intern Aug 2021 – Oct 2021 Punjab Information Technology Board Lahore, Pakistan • Developed a machine learning model to track illegal license plates via CCTV using R. • Collaborated in a team to implement and test the model in a real-world setting. 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EDUCATION UNIVERSITY OF BRITISH COLUMBIA VANCOUVER, BC BACHELOR OF SCIENCE IN COMPUTER SCIENCE SEP. 2023 – MAY 2027 • DEAN'S LIST (2024) EXPERIENCE MACHINE LEARNING INTERN AUG 2021 – OCT 2021 PUNJAB INFORMATION TECHNOLOGY BOARD LAHORE, PAKISTAN • DEVELOPED A MACHINE LEARNING MODEL TO TRACK ILLEGAL LICENSE PLATES VIA CCTV USING R . • COLLABORATED IN A TEAM TO IMPLEMENT AND TEST THE MODEL IN A REAL-WORLD SETTING.

MACHINE LEARNING INTERN JUN 2022 – JUL 2022 HIMMAH CAPITAL LIMITED DUBAI, UAE • APPLIED MACHINE LEARNING TECHNIQUES TO ANALYZE TRENDS IN THE SAUDI STOCK EXCHANGE (TASI). • ENHANCED PREDICTION ACCURACY BY 127% THROUGH KNN AND BIG DATA ANALYTICS . PROJECTS TICKET BOOKING APPLICATION | JAVA, SPRING BOOT, SPRING SECURITY, REACT, VITE • DESIGNED AND IMPLEMENTED A ROBUST TICKET BOOKING PLATFORM WITH SECURE LOGIN AND REGISTRATION FUNCTIONALITY, FEATURING A MODERN, RESPONSIVE UI BUILT WITH REACT AND VITE . • UTILIZED SPRING DATA JPA TO STREAMLINE CRUD OPERATIONS FOR MANAGING USER BOOKINGS, CANCELLATIONS, AND UPDATES, ENSURING SEAMLESS INTERACTION WITH THE BACKEND DATABASE. • IMPLEMENTED SPRING SECURITY AND JWT-BASED AUTHENTICATION TO SAFEGUARD USER DATA AND RESTRICT ACCESS TO AUTHORIZED ENDPOINTS, ENHANCING OVERALL APPLICATION SECURITY. • DEVELOPED RESTful API ENDPOINTS USING SPRING BOOT REST CONTROLLERS , ENABLING EFFICIENT COMMUNICATION BETWEEN THE FRONTEND AND BACKEND. • INTRODUCED A WELL-STRUCTURED SERVICE LAYER TO HANDLE BUSINESS LOGIC, ENSURING SCALABILITY, MAINTAINABILITY, AND SEPARATION OF CONCERNS WITHIN THE APPLICATION. • BACKEND REPOSITORY | FRONTEND REPOSITORY | LIVE APPLICATION FUT TEAM BUILDER APPLICATION | JAVA, SWING, JUNIT, MOCKITO • DEVELOPED A COMPREHENSIVE APPLICATION FOR FIFA ULTIMATE TEAM (FUT) ENTHUSIASTS TO CREATE, MANAGE, AND OPTIMIZE CUSTOM TEAMS WITH AN INTERACTIVE GAMING-STYLE INTERFACE USING JAVA SWING . • IMPLEMENTED SECURE LOGIN AND REGISTRATION FUNCTIONALITY USING JAVA'S USERMANAGER , ENSURING ROLE-BASED ACCESS CONTROL FOR USERS AND ADMINS. • DESIGNED A COMPLEX SERVICE LAYER WITH ALGORITHMS TO CALCULATE KEY TEAM METRICS SUCH AS CHEMISTRY , AVERAGE RATING , AND TOTAL PRICE , IMPROVING THE USER EXPERIENCE. • INTEGRATED A COMMUNITY FEATURE THAT ALLOWS USERS TO SAVE AND SHARE THEIR TEAMS IN A PUBLIC REPOSITORY, ENABLING OTHERS TO BROWSE, FILTER, AND LIKE POPULAR TEAM BUILDS. • ACHIEVED 90% CODE COVERAGE THROUGH COMPREHENSIVE UNIT TESTING USING JUNIT AND MOCKITO , IMPLEMENTING TEST CASES FOR ALL SERVICE LAYER COMPONENTS AND USER INTERACTIONS. • SOURCE CODE SFTWR LABS WEB APPLICATION | JAVA, SPRING BOOT, NEXT.JS • DESIGNED AND DEVELOPED A FULL-STACK BUSINESS APPLICATION FOR SFTWR LABS, FEATURING A SECURE BACKEND BUILT WITH SPRING BOOT AND A DYNAMIC FRONTEND USING NEXT.JS . • IMPLEMENTED A RESTful API ARCHITECTURE AND UTILIZED SPRING DATA JPA FOR CRUD OPERATIONS, MANAGING DATA THROUGH SPRING REST

ENDPOINTS. • DEVELOPED AN ADMIN PANEL WITH SPRING SECURITY AND INTEGRATED SLACK API FOR AUTOMATED NOTIFICATIONS AND SPRING MAIL FOR EMAIL COMMUNICATIONS. • VISIT [SFTWR LABS](#) | [SOURCE CODE \(BACKEND\)](#) | [SOURCE CODE \(FRONTEND\)](#) TECHNICAL SKILLS
LANGUAGES : JAVA, C, C++, PYTHON SQL, JAVASCRIPT, HTML/CSS FRAMEWORKS : SPRING BOOT, REACT, HIBERNATE TOOLS : LINUX, GIT, REDIS CACHE, AMAZON BEDROCK, OPEN AI API, APACHE KAFKA
